📄 Final Report: Employee Performance and Retention Analysis

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1. Objective

The objective of this project is to analyze employee data to understand key factors influencing employee performance and attrition. The project also aims to build machine learning and deep learning models to predict employee attrition and performance, enabling better decision-making for organizations.

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2. Data Preprocessing

The dataset was loaded using Pandas and cleaned to ensure quality:

- Missing values were handled using forward fill method.
- Duplicate records were removed.
- Data types were verified for consistency.

This step ensured that the dataset was reliable for analysis and modeling.

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3. Exploratory Data Analysis (EDA)

Several statistical and visualization techniques were used:

Statistical Analysis

- Mean, median, mode, variance, and standard deviation were calculated.
- These helped in understanding the distribution and spread of numerical features.

Visual Analysis

- **Pairplot** showed relationships between multiple variables.
- **Correlation Heatmap** identified strong relationships between features like salary and performance.
- **Boxplots** helped detect outliers in numerical columns.

Key Observations

- Salary and performance score showed a noticeable correlation.
- Some features had outliers, indicating variability in employee data.

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4. Probability and Statistical Analysis

- Probability of employee attrition was calculated.
- Department-wise attrition probability revealed variation across departments.
- **Bayes' Theorem** was applied to estimate the probability of attrition given high performance.
- **ANOVA Test** was performed:
    - Result indicated whether performance differs significantly across departments.

Conclusion from Statistics

- Employee behavior varies across departments.
- Performance level influences attrition probability.

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5. Feature Engineering

- Categorical variables like **Attrition** and **Department** were encoded using Label Encoding.
- Numerical features like **Salary** and **Performance Score** were scaled using StandardScaler.

This step improved model performance and ensured uniform feature contribution.

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6. Machine Learning Models

6.1 Attrition Prediction (Classification)

- Model Used: **Random Forest Classifier**
- Dataset split into training and testing sets (80:20)

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix

Findings

- The model performed well in predicting employee attrition.
- Confusion matrix showed balanced classification performance.

6.2 Performance Prediction (Regression)

- Model Used: **Linear Regression**

Evaluation Metrics

- R^2 Score
- Mean Squared Error (MSE)

Findings

- The model was able to reasonably predict performance scores.
- Scatter plot of actual vs predicted values showed a linear trend.

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7. Deep Learning Models

7.1 Performance Prediction (Neural Network)

- A feedforward neural network was built with:
    - Input layer
    - Two hidden layers (ReLU activation)
    - Output layer
- Loss Function: Mean Squared Error

Result

- The model achieved acceptable test loss, indicating good prediction capability.

7.2 Attrition Prediction (Neural Network)

- A classification neural network was built using:
  - Sigmoid activation in output layer
  - Binary cross-entropy loss

Result

- The model achieved good accuracy in predicting attrition.

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8. Data Visualization Insights

- **Line Plot**:
  - Showed performance trends based on years at company.
  - Employees with more experience generally showed stable performance.
- **Bar Chart**:
  - Highlighted department-wise attrition differences.
  - Some departments had higher attrition rates.
- **Scatter Plot**:
  - Showed relationship between salary and performance.
  - Moderate positive correlation observed.

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9. Key Findings

- Salary and experience are important factors affecting performance.
- Attrition varies significantly across departments.
- Lower performance employees are more likely to leave.
- Machine learning models can effectively predict attrition and performance.

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10. Recommendations

- Implement **department-specific retention strategies**.
- Improve **employee engagement programs**.
- Provide **performance-based incentives**.
- Focus on **low-performing employees** with training programs.
- Monitor high-risk groups identified by the models.

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11. Conclusion

This project successfully analyzed employee data and built predictive models for performance and attrition. Both machine learning and deep learning approaches provided valuable insights. The results can help organizations improve employee retention, enhance performance, and make data-driven HR decisions.